Taylor Daniels

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ABOUT

Math PhD with expertise in Analytic Number Theory and experience in Mathematical Physics and programming. Currently learning Machine Learning—my background is helping me make rapid progress. Passionate about making tools that solve problems, and looking for roles in Machine Learning, Simulation, Data Science, and Quant. Analysis and Development.

EDUCATION

Ph.D., **Mathematics**, Purdue Univ., West Lafayette, IN	**GPA**: 3.70	**2015 – 2024**
M.S., **Mathematics**, Univ. of Louisville, Louisville, KY	**GPA**: 4.00	**2014 – 2015**
B.S., **Mathematics**, Univ. of Louisville, Louisville, KY	**GPA**: 3.85	**2009 – 2014**

SKILLS

Programming and Computation: Python, numpy, scipy, Sage, Mathematica, matplotlib, Git, JSON
ML/DS: Scikit-learn, pandas, data collection and cleaning, text parsing, regression and classification
Mathematics: Analytic Number Theory, Integer Partitions, Spectral Geometry, Hyperbolic Geometry
Physics: Gauge Theory, Quantum Mechanics, Lagrangian/Hamiltonian mechanics
Certifications: Summer 2026 Data Science Boot Camp – The Erdős Institute
Soft Skills: Excellent at communicating technical topics, excellent written and verbal communication, sociable, driven
Languages: English – (Native); **Chinese/Mandarin** – (Proficient); **Spanish** – (Proficient)

EXPERIENCE

Visiting Assistant Professor, Purdue Univ. Mathematics **2024 – 2026**
- Lead instructor for 8 sections of 200-level Differential Eqns. and Linear Algebra.

Graduate Teaching/Research Assistant, Purdue Univ. Mathematics **2015 – 2024**
- Instructed & graded courses ranging from Calculus up to graduate-level Complex Analysis.
- Awarded Excellent in Teaching by Purdue Mathematics Department in Fall 2020 during the COVID-19 Pandemic.

Instructor, STARTALK Summer Program, Louisville, KY **Summer 2015**
- Fully planned and taught 4 weeks of 6-hour lessons for a course in Mandarin Chinese for students grade 6-8.

Teaching Assistant, Univ. of Louisville Mathematics **2012 – 2015**
- Tutored and assisted teaching work for undergraduate courses from Algebra to Calculus.

SELECTED PROJECTS

Erdős Institute Certificate in Data Science: [GitHub, ExecSummary, Slides, Video] **Summer 2026**
- Analyzed fair balance and statistical biases in randomly generated Pokémon battles.
- Implemented Javascript tools to reverse-engineer random team generation.
- Developed Python tools to analyze 18,000+ Pokémon Showdown battle logs.
- *Skills used*: Python, Javascript, pandas, Scikit-learn, Data Science, Machine Learning

Doctoral and Professional Research: **2021 – Present**
- Computed and analyzed long-term asymptotic behaviors of 1000's of novel weighted integer-partition sequences.
- Developed parallel computation algorithms to overcome (sub-)exponential search space; computational speed increased 10-fold.
- *Skills used*: Python, GP/Pari, Sage, remote computation

Domain coloring: **2022 – 2024**
- Adapted Python library 'cplot' to produce high-resolution magnitude-phase function plots for collaborators.
- The generated function plots guided theoretical math research, and featured in several academic publications.
- *Skills used*: matplotlib, cplot, Mathematica

Graduate-level Mathematical Physics: **2017 – 2019**
- 18 months of advanced-level study of Gauge Theory and the Vector Bundle framework of physical fields.
- Graduate Quantum Mechanics course at Purdue; independent study Lagrangian/Hamiltonian Mechanics.
- *Skills used*: Gauge Theory, Quantum Mechanics, Topology, Differential Geometry

PUBLICATIONS

Vanishing Coefficients in Products of Quintuple Products, T. Daniels et. al; submitted; arXiv	**2026**
The p*-dissection of a product of quintuple products*, T. Daniels et. al.; submitted; arXiv	**2026**
Vanishings of the Legendre-17 signed partition numbers, T. Daniels; arXiv	**2026**
Legendre-signed partition numbers, T. Daniels, Published in J. Math. Anal. Appl. **542** doi	**2025**
Vanishing coefficients in two q*-series…*, T. Daniels; Published in Res. Number Theory **10**; doi	**2024**
The Asymptotics of Some Signed Partition Numbers (Doctoral Thesis), T. Daniels; doi	**2024**
Bounds on the Möbius-signed partition numbers, T. Daniels; Published in Ramanujan J. **65;** doi	**2024**
Biasymptotics of the Möbius- and Liouville-signed partition numbers, T. Daniels, submitted; arXiv	**2023**